

On the Numerical Integration of the Linear Shape Functions Times the 3-D Green's Function or its Gradient on a Plane Triangle

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Abstract—Formulas for the static and dynamic potentials due to linearly varying source distributions defined on a planar triangle are presented. The general linear variation of the source is represented in terms of three linear basis functions, each one associated with a different vertex of the triangular domain. The singular kernels considered are given by the 3-D Green's function and its gradient, both for the static and dynamic case. In the static case the evaluation of potentials is performed analytically and compact form results are given. In the dynamic case the results are given in a form suited for numerical integration.

I. INTRODUCTION

THE numerical formulation of 3-D electromagnetic problems is customarily based either on integrodifferential equations (IE) or on partial differential equations (PDE). Hybrid approaches can also be used in the analysis of complex, penetrable structures. In these approaches, the problem is modeled with PDE on a limited part of the domain and with IE on the remaining, unlimited part. In particular, one may have to numerically solve surface IE when dealing with impenetrable or piecewise homogeneous penetrable bodies, or also with more complex penetrable bodies studied with a hybrid approach.

To numerically solve surface integral equations by application of the method of moments [1], the geometry is usually subdivided into patches of simple shape on which expansion functions for the scalar and/or vector unknowns of the problem can be defined in a simple way. Since the numerical solution of surface IE requires, on a given patch, the evaluation of multiple integrals having singular kernels for coincident observation and integration points, one usually approximates the geometry by means of planar patches to simplify the integrations required. The most commonly used patches are planar triangular ones.

In this paper we address the problem of integrating on a plane triangle the linear shape functions times the static and dynamic (i.e., time-harmonic) 3-D Green's function

and its gradient. It should be evident that solution of this problem permits one to obtain results also for linear expansion functions on a planar polygonal domain. However, the extension of our results to planar polygonal domains is a simple problem which will be dealt with elsewhere.

The interested reader is referred to [2]–[4] for other important results relative to the evaluation of the most common potential integrals due to sources defined on conventional or curved parametric domains; applications of these results can be found, for instance, in [5]–[8]. In particular, formulas for the 3-D potentials due to linearly varying source distributions defined on planar surfaces with polygonal boundaries have previously appeared in [2]. However, the singular kernels considered in [2] are only relative to Green's functions and not to their gradients; the new results provided in the present paper permit the evaluation of potential integrals having as kernels the gradients of the (static and dynamic) 3-D Green's functions. Although these latter integrals can be evaluated in the principal value sense, after the analytical extraction of the correct residue contributions arising when the observation point is in the source region, they are expressed here in compact form in terms of readily identifiable geometrical quantities, thereby facilitating the numerical implementation. We observe that surface integral formulations are greatly simplified when the evaluation of both the principal value and the residue contribution is performed numerically; in this instance, in fact, there is no more need to analytically extract the residue contributions while developing the formulation.

II. GEOMETRY OF THE INTEGRATION DOMAIN, LINEAR EXPANSION FUNCTIONS AND AUXILIARY INTEGRATION FUNCTIONS

With reference to Fig. 1(a), for the plane triangle T , arbitrarily located in the Cartesian frame $\mathcal{S} \equiv (x, y, z)$, we introduce a local reference frame $\mathcal{S} \equiv (u, v, w)$ having origin on the first vertex of the triangle (node 1); in the local frame, T lies in the plane $w = 0$. The contour ∂T of the triangle is oriented by counterclockwise numbering with respect to $\hat{w}$ the nodes of T , from 1 to 3; the u axis passes through node 2. Let us denote as ∂T_i the side

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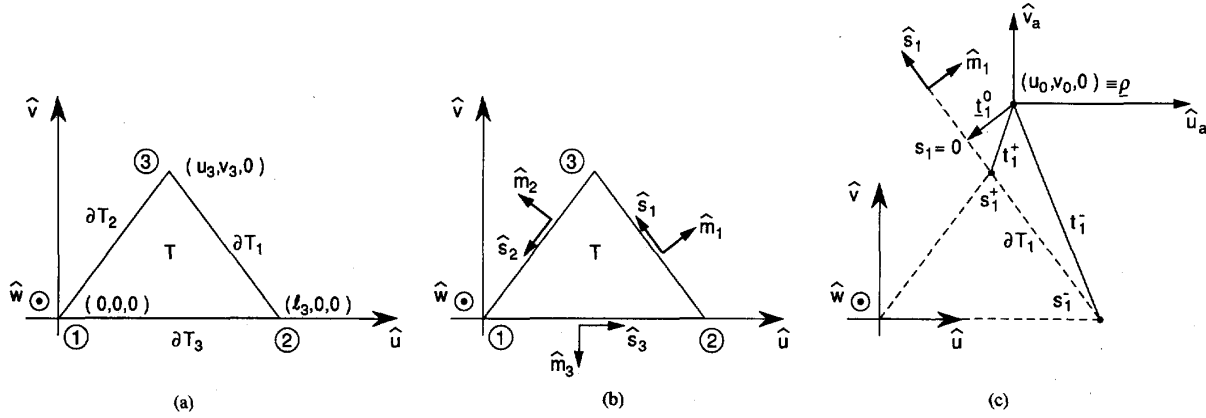


Fig. 1. (a) Plane triangle T in its local frame $\mathcal{S} \equiv (u, v, w)$. The length of the i -th side, ∂T_i , is denoted as l_i . (b) Tangent ($\hat{s}$) and normal ($\hat{m}$) unit vectors on the triangle contour. (c) The source integrals are evaluated in the Cartesian frame $\mathcal{S}_a \equiv (u_a, v_a, w)$, obtained by translating $\mathcal{S}$ to the new origin $\rho \equiv (u_0, v_0, 0)_{\mathcal{S}}$.

opposite to node i ; the side length is l_i . In the local frame, the triangle nodes have the following (u, v) coordinates:

$$\text{node 1} \equiv (0, 0); \quad \text{node 2} \equiv (l_3, 0);$$

$$\text{node 3} \equiv (u_3, v_3).$$

Fig. 1(b) shows, on the three sides of T , the counterclockwise oriented unit vector $\hat{s}$, tangent to ∂T_i ($i = 1, 3$), and the outgoing unit vector $\hat{m} = \hat{s} \times \hat{w}$, perpendicular to ∂T_i . The necessary relations between the local (u, v, w) coordinate system and global coordinates are given in Section 5.

On the triangle, we assume that the electromagnetic (vector or scalar) sources can be represented as linear combinations of the functions N_1, N_2, N_3 , with [9]:

$$N = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & \left(\frac{u_3}{l_3} - 1\right) \\ 0 & 1 & -\frac{u_3}{l_3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{u} \\ \frac{1}{l_3} \\ \frac{v}{v_3} \end{bmatrix}, \quad (1)$$

$$N_1 + N_2 + N_3 = 1, \quad (2)$$

and where N_i is the linear (nodal) function equal to unity at the i -th node and zero at the other two. Our problem is to integrate, for a given observation point $\mathbf{r}$ in the global frame $\mathcal{S}$, the source distribution (i.e., the nodal functions) times the 3-D scalar Green's function and its gradient. The integration point in T is denoted as $\mathbf{r}'$, while the vector distance from observation to integration point is $\mathbf{R} = \mathbf{r} - \mathbf{r}'$. In the local frame $\mathcal{S}$, one can write $\mathbf{r} = \rho + w_0 \hat{w} \equiv (u_0, v_0, w_0)$ and $\mathbf{r}' \equiv (u', v', 0)$.

For the i -th side, we will use the superscript $-$, $+$ to denote quantities related to the first and second node of ∂T_i , respectively. With reference to Fig. 1(c), the point ρ (i.e., the projection of the observation point $\mathbf{r}$ on the plane of the triangle: $\rho = (\hat{w} \times \mathbf{r}) \times \hat{w}$) permits one to

define on ∂T_i the arclength variable s_i , increasing in the $\hat{s}_i$ direction from the value s_i^- at the first node to the final value $s_i^+ = s_i^- + l_i$ at the second node; the origin $s_i = 0$ individuates the perpendicular projection of ρ on ∂T_i (or its extension). At the endpoints of ∂T_i the s the coordinate takes the following values:

$$\begin{aligned} s_1^- &= -\frac{(l_3 - u_0)(l_3 - u_3) + v_0 v_3}{l_1}, \\ s_1^+ &= \frac{(u_3 - u_0)(u_3 - l_3) + v_3(v_3 - v_0)}{l_1}, \\ s_2^- &= -\frac{u_3(u_3 - u_0) + v_3(v_3 - v_0)}{l_2}, \\ s_2^+ &= \frac{u_0 u_3 + v_0 v_3}{l_2}, \\ s_3^- &= -u_0, \quad s_3^+ = l_3 - u_0 \end{aligned} \quad (3)$$

Let us call $\mathbf{t}$ the distance vector from ρ to a point in the plane of the triangle; in Fig. 1(c) we show the vector $\mathbf{t}_i^0 = t_i^0 \hat{m}_i$ drawn from the point ρ to the point $s_i = 0$ of the i -th side, and the two distances t_i^-, t_i^+ from ρ to the two nodes of ∂T_i . For these quantities, one has:

$$\begin{aligned} t_1^0 &= \frac{v_0(u_3 - l_3) + v_3(l_3 - u_0)}{l_1}, \\ t_2^0 &= \frac{u_0 v_3 - v_0 u_3}{l_2}, \\ t_3^0 &= v_0; \\ t_3^+ &= t_1^-, \quad t_2^- = t_1^+, \quad t_3^- = t_2^+, \\ t_1^- &= \sqrt{(l_3 - u_0)^2 + v_0^2}, \\ t_1^+ &= \sqrt{(u_3 - u_0)^2 + (v_3 - v_0)^2}, \\ t_2^+ &= \sqrt{u_0^2 + v_0^2}. \end{aligned} \quad (4)$$

$$t_2^+ = \sqrt{u_0^2 + v_0^2}. \quad (5)$$

In the following, $R_i^0 = \sqrt{(t_i^0)^2 + w_0^2}$ is the distance from the observation point $\mathbf{r}$ to the i -th side, whereas R_i^- and R_i^+ are the distances from the observation point to the endpoints of ∂T_i .

To perform the source integrals, it is convenient to translate $\mathcal{L}$ into the new Cartesian frame $\mathcal{L}_a \equiv (u_a, v_a, w)$ having origin at $\boldsymbol{\rho}$ (see Fig. 1(c)). The integrals are first evaluated for $w_0 \neq 0$; the results for observation points lying in the plane of the triangle are then found in the limit for $w_0 = 0$.

Evaluation of the integrals is performed by using appropriate Gauss integral theorems to transform the integration on T into one integration over the boundary ∂T of T . This procedure requires that the integrands are continuously differentiable on the integration domain and therefore, when the point $\boldsymbol{\rho} \equiv (u_0, v_0, 0)_{\mathcal{L}}$ is located in T or on its contour, one may have to break T into the two regions T_e and $(T - T_e)$. T_e is the intersection of T and a small disk of radius ϵ centered at $\boldsymbol{\rho}$. The integrals over T_e or its boundary, ∂T_e , require a separate treatment in a local polar coordinate system having origin at $\boldsymbol{\rho}$. Such integrals involve the angular extent $\alpha(\boldsymbol{\rho})$ of the circular arc portion of ∂T_e lying within T [2]. To this purpose, one has to distinguish between the four different cases (see [2, Fig. 3]):

1. $\alpha(\boldsymbol{\rho}) = 0$, if $\boldsymbol{\rho}$ is outside T ;
2. $\alpha(\boldsymbol{\rho}) = 2\pi$, if $\boldsymbol{\rho}$ is inside T ;
3. $\alpha(\boldsymbol{\rho}) = \pi$, if $\boldsymbol{\rho}$ is on ∂T but not at a corner;
4. $\alpha(\boldsymbol{\rho}) = \alpha$, if $\boldsymbol{\rho}$ is at a vertex of T , α being the angle between the two edges of T meeting at the vertex ($0 < \alpha < \pi$).

To integrate in the $\mathcal{L}_a$ frame, the parametric representation used for the i -th side is as follows (see Fig. 1(c)):

$$\begin{bmatrix} u_a(s_i) \\ v_a(s_i) \end{bmatrix} = (s_i \hat{s}_i + t_i^0) \cdot \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix}; \quad (6)$$

then, in order to use Gauss integral theorems in $\mathcal{L}_a$, we further observe that the following relationships hold (see [4, eq. 47]):

$$\begin{bmatrix} u'_a/R \\ v'_a/R \end{bmatrix} = \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} \cdot \nabla'_s R, \quad (7)$$

$$\frac{1}{R^3} = -\nabla'_s \cdot \frac{\mathbf{t}}{t^2 R}, \quad (8)$$

$$\begin{bmatrix} u'_a \\ v'_a \end{bmatrix} \frac{1}{R^3} = -\nabla'_s \cdot \begin{bmatrix} \hat{u}/R \\ \hat{v}/R \end{bmatrix}, \quad (9)$$

$$\begin{bmatrix} u'_a \\ v'_a \end{bmatrix} \nabla'_s \frac{1}{R} = \nabla'_s \begin{bmatrix} u'_a/R \\ v'_a/R \end{bmatrix} - \frac{1}{R} \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix}. \quad (10)$$

In (7)–(10), the surface gradient and divergence operators that operate on the primed (integration) coordinates are denoted as ∇'_s and $(\nabla'_s \cdot)$, respectively.

In the following section, we will express the results as summation of *side-functions* of the observation point; in

particular, to express results concisely, we will use the same functions introduced in [4, eq. 51]:

$$f_{2i} = \ln \left(\frac{R_i^+ + s_i^+}{R_i^- + s_i^-} \right), \quad (11)$$

$$f_{3i} = (s_i^+ R_i^+ - s_i^- R_i^-) + (R_i^0)^2 f_{2i}; \quad (12)$$

plus the auxiliary functions (see [2, eq. 5]):

$$\beta_i = \tan^{-1} \frac{t_i^0 s_i^+}{(R_i^0)^2 + |w_0| R_i^+} - \tan^{-1} \frac{t_i^0 s_i^-}{(R_i^0)^2 + |w_0| R_i^-}, \quad (13)$$

with

$$\beta = \sum_{i=1}^3 \beta_i. \quad (14)$$

In (13), each arctangent function is evaluated on its principal branch and t_i^0 has positive sign if $\mathbf{t}_i^0$ and $\hat{\mathbf{m}}_i$ are pointing in the same direction; it has negative sign if $\mathbf{t}_i^0$ and $\hat{\mathbf{m}}_i$ have opposite directions (see (4)). If the observation point is on the i -th edge of T or its extension (i.e., when $t_i^0 = 0$), one has $\beta_i = 0$.

In the limit of observation point lying in the plane of the triangle one obtains:

$$\hat{f}_{2i} = f_{2i}(w_0 = 0) = \ln \left(\frac{t_i^+ + s_i^+}{t_i^- + s_i^-} \right), \quad (15)$$

$$\hat{f}_{3i} = f_{3i}(w_0 = 0) = (s_i^+ t_i^+ - s_i^- t_i^-) + (t_i^0)^2 \hat{f}_{2i}, \quad (16)$$

$$\hat{\beta}_i = \beta_i(w_0 = 0) = \tan^{-1} \frac{s_i^+}{t_i^0} - \tan^{-1} \frac{s_i^-}{t_i^0}; \quad (17)$$

with

$$\alpha(\boldsymbol{\rho}) = \sum_{i=1}^3 \hat{\beta}_i. \quad (18)$$

Note that, to be consistent with (13), in (17) we set $\hat{\beta}_i = 0$ at $t_i^0 = 0$, by definition (i.e., $\{\beta_i = 0 \text{ if } t_i^0 = 0, \forall w_0\}$).

III. STATIC POTENTIAL INTEGRALS

A. Integrals having kernel $1/R$

Let us call $\mathbf{R} = \mathbf{r} - \mathbf{r}'$ the vector distance from the observation point $\mathbf{r} \equiv (u_0, v_0, w_0)$ and the integration point $\mathbf{r}' \equiv (u', v', 0)$, located on T ; with $R = |\mathbf{R}|$. The integral over T of the potential function $1/R$ can be easily obtained from [2, eq. 5]; according to our notations, one has

$$\begin{aligned} I_1 &= \int_T \frac{1}{R} dT' = \sum_{i=1}^3 (t_i^0 f_{2i} - |w_0| \beta_i) \\ &= -|w_0| \beta + \sum_{i=1}^3 t_i^0 f_{2i}. \end{aligned} \quad (19)$$

When the observation point is on an edge of T or its extension (i.e., when $t_i^0 = 0$), (13) permits one to easily verify that the contribution to the sum in (19) from that edge vanishes.

Since R is continuously differentiable on the integration domain for $w_0 \neq 0$, (7) along with [10, eq. (230.01)] yield

$$\begin{aligned} \begin{bmatrix} I_{u_a} \\ I_{v_a} \end{bmatrix} &= \int_T \begin{bmatrix} u'_a \\ v'_a \end{bmatrix} \frac{1}{R} dT' = \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} \cdot \int_T \nabla'_s R dT' \\ &= \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} \cdot \int_{\partial T} \hat{m} R ds' = \frac{1}{2} \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} \cdot \sum_{i=1}^3 \hat{m}_i f_{3i}, \end{aligned} \quad (20)$$

which, as a special case, coincides with the result reported in [2, eq. 6]. Note that (20) are valid also in the limit $w_0 = 0$, i.e., for observation point lying in the plane of the triangle.

By recalling that $u' = u'_a + u_0$ and $v' = v'_a + v_0$, the previous results yield

$$\begin{bmatrix} I_u \\ I_v \end{bmatrix} = \int_T \begin{bmatrix} u' \\ v' \end{bmatrix} \frac{1}{R} dT' = \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} I_1 + \begin{bmatrix} I_{u_a} \\ I_{v_a} \end{bmatrix}. \quad (21)$$

When the observation point lies in the plane of the triangle, (19) and (20) simplify as follows ($w_0 = 0$);

$$I_1 = \sum_{i=1}^3 t_i^0 f_{2i}^0, \quad (22)$$

$$\begin{bmatrix} I_{u_a} \\ I_{v_a} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} \cdot \sum_{i=1}^3 \hat{m}_i f_{3i}^0. \quad (23)$$

The previous results together with the expressions of the linear nodal functions (1) immediately yield

$$\begin{aligned} \int_T N \frac{1}{R} dT' &= \begin{bmatrix} 1 & -1 & \left(\frac{u_3}{l_3} - 1\right) \\ 0 & 1 & -\frac{u_3}{l_3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_u \\ I_v \end{bmatrix} \\ &= \begin{bmatrix} N_{10} \\ N_{20} \\ N_{30} \end{bmatrix} I_1 + \begin{bmatrix} -1 & \left(\frac{u_3}{l_2} - 1\right) \\ 1 & -\frac{u_3}{l_3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I_{u_a} \\ I_{v_a} \\ v_3 \end{bmatrix}, \end{aligned} \quad (24)$$

where N_{i0} ($i = 1, 3$) is the value reached by the i -th nodal function at the point (u_0, v_0) .

B. Integrals having kernel $\nabla(R^{-1})$

In order to integrate linear functions of u, v times the gradient of the potential function $1/R$ it is convenient to separate $\nabla(R^{-1})$ into two terms:

$$\nabla \frac{1}{R} = -\nabla' \frac{1}{R} = -\left(\hat{w} \frac{w_0}{R^3} + \nabla'_s \frac{1}{R} \right), \quad (25)$$

where ∇', ∇'_s , operate on the primed coordinates, whereas ∇ operates on the unprimed (observation) coordinates. The contributions to potential integrals due to the first term in parentheses of (25) are evaluated with the aid of [10, eqs. (200.01.), (201.03.), (387.)] and of (8)–(9) as follows ($w_0 \neq 0$):¹

$$\begin{aligned} w_0 \int_T \frac{dT'}{R^3} &= w_0 \lim_{\epsilon \rightarrow 0} \left\{ \int_{T_\epsilon} \frac{dT'}{R^3} - \int_{T-T_\epsilon} \nabla'_s \cdot \frac{\mathbf{t}}{t^2 R} dT' \right\} \\ &= w_0 \lim_{\epsilon \rightarrow 0} \left\{ \int_{T_\epsilon} \frac{dT'}{R^3} - \int_{\partial(T-T_\epsilon)} \frac{\hat{m} \cdot \mathbf{t}}{t^2 R} ds' \right\} \\ &= w_0 \left\{ \lim_{\epsilon \rightarrow 0} \left[\int_{T_\epsilon} \frac{dT'}{R^3} + \int_\alpha \frac{d\phi'}{R} \right] - \int_{\partial T} \frac{\hat{m} \cdot \mathbf{t}}{t^2 R} ds' \right\} \\ &= w_0 \left\{ \frac{\alpha(\rho)}{|w_0|} - \int_{\partial T} \frac{\hat{m} \cdot \mathbf{t}}{t^2 R} ds' \right\} \\ &= \text{sgn}(w_0) \left\{ \alpha(\rho) - \sum_{i=1}^3 \left(\tan^{-1} \frac{s_i^+ |w_0|}{t_i^0 R_i^+} \right. \right. \\ &\quad \left. \left. - \tan^{-1} \frac{s_i^- |w_0|}{t_i^0 R_i^-} \right) \right\} \\ &= \text{sgn}(w_0) \sum_{i=1}^3 \beta_i = \text{sgn}(w_0) \beta. \end{aligned} \quad (26)$$

$$\begin{aligned} w_0 \int_T \begin{bmatrix} u'_a \\ v'_a \end{bmatrix} \frac{dT'}{R^3} &= -w_0 \int_T \nabla'_s \cdot \begin{bmatrix} \hat{u}/R \\ \hat{v}/R \end{bmatrix} dT' \\ &= -w_0 \int_{\partial T} \frac{\hat{m}}{R} \cdot \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} ds' \\ &= -w_0 \sum_{i=1}^3 \hat{m}_i \cdot \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} f_{2i}. \end{aligned} \quad (27)$$

In (26), the summation of the angles β_i properly takes into account the residue contribution involving the angle α of the portion of ∂T_ϵ in T . This result has been obtained by expressing α as a sum of angles between adjacent vertices around the triangle (see (17) and (18)), thereby combining the residue contribution with the finite

¹To further clarify (26) and the origin of the residue contribution involving the angle α of the circular arc portion of ∂T_ϵ lying within T , notice the following results ($w_0 \neq 0$):

$$\begin{aligned} \int_{T_\epsilon} \frac{dT'}{R^3} - \int_\alpha \frac{\hat{m} \cdot \mathbf{t}_\epsilon}{t_\epsilon^2 R} d\phi' &= \left[\frac{\alpha(\rho)}{|w_0|} - \frac{\alpha(\rho)}{\sqrt{\epsilon^2 + w_0^2}} \right] + \int_\alpha \frac{d\phi'}{R} = \frac{\alpha(\rho)}{|w_0|}, \\ \lim_{\epsilon \rightarrow 0} \int_{T_\epsilon} \frac{dT'}{R^3} &= \frac{\alpha(\rho)}{|w_0|} - \lim_{\epsilon \rightarrow 0} \int_\alpha \frac{d\phi'}{R} = 0. \end{aligned}$$

summation (see the discussion following [2, eq. (5)]). In the limit $w_0 = 0$ (26) and (27) yield

$$\lim_{w_0 \rightarrow 0} \left(-\hat{w} w_0 \int_T \frac{dT'}{R^3} \right) = \pm \hat{w} \sum_{i=1}^3 \hat{\beta}_i = \pm \alpha(\rho) \hat{w}; \quad (28)$$

$$\lim_{w_0 \rightarrow 0} w_0 \int_T \left[\frac{u'_a}{v'_a} \right] \frac{dT'}{R^3} = 0. \quad (29)$$

In (28), the plus and minus signs correspond to an approach from negative and from positive values of w_0 , respectively.

To consider contributions to potential integrals due to the factor $\nabla'_s(1/R)$ (i.e., the second term in parentheses of (25)), use of (6), (10), and (19) with the aid of [10, eqs. (200.01.), (201.01.)] yield

$$\begin{aligned} \int_T \nabla'_s \frac{1}{R} dT' &= \int_{\partial T} \frac{\hat{m}}{R} ds' = \sum_{i=1}^3 \hat{m}_i f_{2i}, \\ - \int_T \left[\frac{u'_a}{v'_a} \right] \nabla'_s \frac{1}{R} dT' &= \left[\frac{\hat{u}}{\hat{v}} \right] \int_T \frac{1}{R} dT' - \int_{\partial T} \hat{m} \left[\frac{u'_a/R}{v'_a/R} \right] ds' \\ &= \sum_{i=1}^3 \left\{ \left[\frac{\hat{u}}{\hat{v}} \right] (t_i^0 f_{2i} - |w_0| \beta_i) - \hat{m}_i \left[\frac{\hat{u}}{\hat{v}} \right] \right. \\ &\quad \cdot \left. \left[\hat{s}_i (R_i^+ - R_i^-) + t_i^0 f_{2i} \right] \right\} \\ &= - \left[\frac{\hat{u}}{\hat{v}} \right] |w_0| \beta + \sum_{i=1}^3 f_i \left[\frac{\hat{u} \cdot \hat{s}_i}{\hat{v} \cdot \hat{s}_i} \right]; \end{aligned} \quad (30)$$

with

$$f_i = \hat{s}_i t_i^0 f_{2i} - \hat{m}_i (R_i^+ - R_i^-), \quad (32)$$

and where the following intermediate result has been used in deriving (31):

$$\int_{\partial T} \left[\frac{u'_a/R}{v'_a/R} \right] ds' = \left[\frac{\hat{u}}{\hat{v}} \right] \cdot \sum_{i=1}^3 \{ \hat{s}_i (R_i^+ - R_i^-) + t_i^0 f_{2i} \}. \quad (33)$$

By recalling that $u' = u'_a + u_0$ and $v' = v'_a + v_0$, (25)–(32) yield ($w_0 \neq 0$)

$$I_1^\nabla = \int_T \nabla \frac{1}{R} dT' = -\hat{w} \operatorname{sgn}(w_0) \beta - \sum_{i=1}^3 \hat{m}_i f_{2i}, \quad (34)$$

$$\begin{bmatrix} I_{u_a}^\nabla \\ I_{v_a}^\nabla \end{bmatrix} = \int_T \left[\frac{u'}{v'} \right] \nabla \frac{1}{R} dT' = \begin{bmatrix} u_0 \\ v_0 \end{bmatrix} I_1^\nabla + \begin{bmatrix} I_{u_a}^\nabla \\ I_{v_a}^\nabla \end{bmatrix}, \quad (35)$$

$$\begin{aligned} \begin{bmatrix} I_{u_a}^\nabla \\ I_{v_a}^\nabla \end{bmatrix} &= \int_T \left[\frac{u'_a}{v'_a} \right] \nabla \frac{1}{R} dT' \\ &= \hat{w} w_0 \left[\frac{\hat{u}}{\hat{v}} \right] \cdot \sum_{i=1}^3 \hat{m}_i f_{2i} - \left[\frac{\hat{u}}{\hat{v}} \right] |w_0| \beta + \sum_{i=1}^3 f_i \left[\frac{\hat{u} \cdot \hat{s}_i}{\hat{v} \cdot \hat{s}_i} \right]. \end{aligned} \quad (36)$$

The previous results are valid also in the limit $w_0 = 0$; in this particular case one obtains

$$I_1^\nabla = \pm \hat{w} \alpha(\rho) - \sum_{i=1}^3 \hat{m}_i f_{2i}, \quad (37)$$

$$\begin{bmatrix} I_{u_a}^\nabla \\ I_{v_a}^\nabla \end{bmatrix} = \sum_{i=1}^3 f_i \begin{bmatrix} \hat{u} \cdot \hat{s}_i \\ \hat{v} \cdot \hat{s}_i \end{bmatrix}, \quad (38)$$

$$f_i = \hat{s}_i t_i^0 f_{2i} - \hat{m}_i (t_i^+ - t_i^-). \quad (39)$$

In (37), the plus and minus signs correspond to an approach from negative and from positive values of w_0 , respectively.

The previous results together with the expressions of the linear nodal functions (1) yield

$$\int_T N \nabla \left(\frac{1}{R} \right) dT' = \begin{bmatrix} N_{10} \\ N_{20} \\ N_{30} \end{bmatrix} I_1^\nabla + \begin{bmatrix} -1 & \left(\frac{u_3}{l_3} - 1 \right) \\ 1 & -\frac{u_3}{l_3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} I_{u_a}^\nabla \\ I_{v_a}^\nabla \\ v_3 \end{bmatrix}. \quad (40)$$

Equation (40) is valid for all w_0 ; at $w_0 = 0$ (and $\rho \equiv (u_0, v_0) \in T$), the right-hand side of (40) is obtained from (37)–(39), whereas the left-hand side may be rewritten in terms of a surface integral in the principal value sense plus a residue contribution;

$$\int_T N \nabla \left(\frac{1}{R} \right) dT' = \oint_T N \nabla \left(\frac{1}{R} \right) dT' \pm \hat{w} \alpha(\rho) N(\rho). \quad (41)$$

IV. DYNAMIC POTENTIAL INTEGRALS

The results of the previous section permit the numerical evaluation of the moment matrix elements, as required by application of the method of moments to solve electromagnetic dynamic problems modeled by using surface triangular patches. The scalar Green's function relative to dynamic 3-D problems is $G = \exp(-jkR)/R$, where k is the wavenumber associated with the assumed $\exp(j\omega t)$ time variation. To deal with dynamic problems it is often convenient to introduce the normalized coordinate frame $\tilde{r} = k\mathbf{r}$ [7], [8], where the scalar Green's function reads

$$G = k \frac{\exp(-j\tilde{R})}{\tilde{R}}, \quad (42)$$

with

$$\tilde{\mathbf{R}} = \tilde{\mathbf{r}} - \tilde{\mathbf{r}}', \quad \tilde{R} = |\tilde{\mathbf{R}}|, \quad \hat{\mathbf{R}} = \frac{\tilde{\mathbf{R}}}{\tilde{R}}. \quad (43)$$

The dynamic potential integral relative to the linear nodal function N_i ($i = 1, 3$) can be expressed as follows

$$\begin{aligned} k \int_T N_i(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dT' &= \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') G(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}') d\tilde{T}' \\ &= I_{\text{anl}} + I_{\text{num}}, \end{aligned} \quad (44)$$

with

$$I_{\text{anl}} = \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \frac{1}{\tilde{R}} d\tilde{T}', \quad (45)$$

$$I_{\text{num}} = \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \frac{\exp(-j\tilde{R}) - 1}{\tilde{R}} d\tilde{T}'. \quad (46)$$

The integral (45) is readily evaluated by use of (24); the second integral (46) can be integrated numerically, since its integrand is bounded for every observation point.²

Similarly, when the kernel is the gradient of the scalar Green's function, one may break the potential integral into two integrals: I_{anl}^{∇} , evaluated analytically, and I_{num}^{∇} , to be integrated numerically

$$\begin{aligned} \int_T N_i(\mathbf{r}') \nabla G(\mathbf{r}, \mathbf{r}') dT' &= \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} G(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}') d\tilde{T}' \\ &= I_{\text{anl}}^{\nabla} + I_{\text{num}}^{\nabla}, \end{aligned} \quad (47)$$

where $\nabla_{\tilde{\mathbf{r}}}$ operates on the *normalized* observation coordinates. For observation points located outside the integration triangle, one chooses

$$I_{\text{anl}}^{\nabla} = \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left(\frac{1}{\tilde{R}} \right) d\tilde{T}', \quad (48)$$

$$I_{\text{num}}^{\nabla} = \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left[\frac{\exp(-j\tilde{R}) - 1}{\tilde{R}} \right] d\tilde{T}'; \quad (49)$$

with I_{anl}^{∇} evaluated by use of (40).

The integrand of (49) is always bounded, but discontinuous at $\tilde{R} = 0$; in fact, for small values of $\tilde{R}$ (i.e., for the observation point close to the integration point) one has

$$N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left[\frac{\exp(-j\tilde{R}) - 1}{\tilde{R}} \right] \approx -N_{i0} \frac{\hat{\mathbf{R}}}{2}. \quad (50)$$

Thus, to ease the numerical integration process when the observation point is inside the integration region T , it is convenient to choose.³

²In fact:

$$\lim_{\tilde{R} \rightarrow 0} \frac{\exp(-j\tilde{R}) - 1}{\tilde{R}} = -j.$$

³For small values of $\tilde{R}$ the integrand of (51), which is continuous and equal to zero at $\tilde{R} = 0$, is numerically approximated by truncation of its series representation,

$$\left\{ \frac{N_{i0} - N_i(\tilde{\mathbf{r}}')}{2} - N_i(\tilde{\mathbf{r}}') \sum_{p=1}^{\infty} \frac{(-j\tilde{R})^p}{p!(p+2)} \right\} \hat{\mathbf{R}}.$$

$$I_{\text{num}}^{\nabla} = \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left[\frac{\exp(-j\tilde{R}) - 1}{\tilde{R}} \right] + N_{i0} \frac{\hat{\mathbf{R}}}{2} d\tilde{T}', \quad (51)$$

which, by use of (47) and (20) (or [2, eq. (6)]), yield

$$\begin{aligned} I_{\text{anl}}^{\nabla} &= \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left(\frac{1}{\tilde{R}} \right) d\tilde{T}' - \frac{N_{i0}}{2} \int_{\tilde{T}} \hat{\mathbf{R}} d\tilde{T}' \\ &= \int_{\tilde{T}} N_i(\tilde{\mathbf{r}}') \nabla_{\tilde{\mathbf{r}}} \left(\frac{1}{\tilde{R}} \right) d\tilde{T}' + \frac{N_{i0}}{4} \sum_{i=1}^3 \hat{\mathbf{m}}_i f_{3i}. \end{aligned} \quad (52)$$

Obviously, one has to work with normalized coordinates in order to use the result (40) and to compute the auxiliary function f_{3i} , as required by (52).

V. NUMERICAL INTEGRATION TECHNIQUE

Let us consider the integration triangle T of area Δ in the global frame $\mathcal{S} \equiv (x, y, z)$, where the vertices of T are $\mathbf{P}_i = (x_i, y_i, z_i)$ ($i = 1, 3$). The following discussion also holds for the normalized coordinate case. By counter-clockwise ordering the vertices with respect to the normal vector⁴

$$\mathbf{n} = (\mathbf{P}_2 - \mathbf{P}_1) \times (\mathbf{P}_3 - \mathbf{P}_1), \quad (53)$$

one has

$$|\mathbf{n}| = 2\Delta, \quad \hat{\mathbf{n}} = \frac{\mathbf{n}}{2\Delta}; \quad (54)$$

$$l_i = |\mathbf{P}_{i-1} - \mathbf{P}_{i+1}|, \quad (i = 1, 3),$$

$$\hat{\mathbf{s}}_i = \frac{\mathbf{P}_{i-1} - \mathbf{P}_{i+1}}{l_i}, \quad \sum_{i=1}^3 l_i \hat{\mathbf{s}}_i = 0,$$

$$\hat{\mathbf{m}}_i = \hat{\mathbf{s}}_i \times \hat{\mathbf{n}}, \quad \sum_{i=1}^3 l_i \hat{\mathbf{m}}_i = 0; \quad (55)$$

with $(i+1) = 1$ for $i = 3$, and $(i-1) = 3$ for $i = 1$. In terms of global coordinates, the unit vectors $(\hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\mathbf{w}})$ are

$$\hat{\mathbf{u}} = \frac{\mathbf{P}_2 - \mathbf{P}_1}{l_3}, \quad \hat{\mathbf{v}} = \hat{\mathbf{n}} \times \frac{\mathbf{P}_2 - \mathbf{P}_1}{l_3}, \quad \hat{\mathbf{w}} = \hat{\mathbf{n}}, \quad (56)$$

with

$$\begin{bmatrix} \hat{\mathbf{u}} \cdot \hat{\mathbf{s}}_i \\ \hat{\mathbf{v}} \cdot \hat{\mathbf{s}}_i \end{bmatrix} = \begin{bmatrix} -\hat{\mathbf{v}} \cdot \hat{\mathbf{m}}_i \\ \hat{\mathbf{u}} \cdot \hat{\mathbf{m}}_i \end{bmatrix}; \quad (57)$$

while the $\mathcal{L}$ -coordinates of the third node and of the observation point $\mathbf{r}$ of the global frame are given as follows,

$$u_3 = (\mathbf{P}_3 - \mathbf{P}_1) \cdot \hat{\mathbf{u}}, \quad v_3 = \frac{2\Delta}{l_3}. \quad (58)$$

$$\begin{bmatrix} u_0 \\ v_0 \\ w_0 \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{u}} \\ \hat{\mathbf{v}} \\ \hat{\mathbf{w}} \end{bmatrix} \cdot (\mathbf{r} - \mathbf{P}_1). \quad (59)$$

⁴For example, $\hat{\mathbf{n}}$ is the outgoing (or the ingoing) unit normal for all the triangular patches defined on a closed (conducting) surface.

Equations (53)–(59) permit one to evaluate the integrals of Section III directly in global coordinates.

As far as the numerical integrals of Section IV are concerned, it is convenient to work with triangle area coordinates (η, ξ) [9]:

$$\begin{bmatrix} \eta \\ \xi \end{bmatrix} = \frac{1}{2\Delta} \begin{bmatrix} v_3 & -u_3 \\ 0 & l_3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}, \quad \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} l_3 & u_3 \\ 0 & v_3 \end{bmatrix} \begin{bmatrix} \eta \\ \xi \end{bmatrix}. \quad (60)$$

In fact, in the (η, ξ) parent space the integration triangle is mapped by the simplex:

$$\{0 \leq \eta \leq 1; 0 \leq \xi \leq 1 - \eta\}; \quad (61)$$

the linear shape functions are

$$\begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \eta \\ \xi \end{bmatrix}; \quad (62)$$

while the global coordinates of the integration point $\mathbf{r}_s = \mathbf{r}'$ are easily obtained as follows,

$$\mathbf{r}_s \equiv [x_s, y_s, z_s] = [N_1(\eta_s, \xi_s), N_2(\eta_s, \xi_s), N_3(\eta_s, \xi_s)] \cdot \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{bmatrix}. \quad (63)$$

For a given integration point in (η, ξ) , (63) permits one to evaluate the vector distance from observation and integration point, in the global space; therefore, the kernels of (46), (49), and (51) can be easily evaluated to provide integral results directly in global coordinates.

In Table I we report, for completeness, the set of points (η_s, ξ_s) , and their relative weights W_s , to numerically evaluate the dynamic potential integrals of Section IV when the observation point is located inside the integration triangle (i.e., self-element case). The integration formula of Table I is the seven point formula of fifth degree reported in [11, p. 314]. This formula has been used to integrate on triangles having maximum dimension of the order of $1/3$ of a wavelength; however, note that appropriate application of the method of moments with linear expansion functions requires patches of smaller dimension.

For self-element evaluation, unless one uses a delta-sampling technique, one needs to perform a second *observation* (or testing) integral on the same triangle where the sources are defined. This integral can be performed with the integration formula reported in Table II, which has been derived from the three point formula of second degree given in [11, p. 307], by dividing the parent simplex into three triangles of equal area. The weights of Tables I and II are proportional to the area Δ of the integration triangle T . The integration points of Tables I and II never coincide; they are located inside T and not on the bound-

TABLE I
SEVEN POINT INTEGRATION FORMULA

Sample η_s, ξ_s	v_1, v_1	v_2, v_2	v_2, v_3	v_3, v_2	v_4, v_4	v_4, v_5	v_5, v_4
Weight	W_a	W_b	W_b	W_b	W_c	W_c	W_c
	$v_1 = \frac{1}{3}, \quad W_a = \frac{9}{40}\Delta,$						
	$v_2 = \frac{6 - \sqrt{15}}{21},$	$v_3 = \frac{9 + 2\sqrt{15}}{21},$	$W_b = \frac{155 - \sqrt{15}}{1200}\Delta,$				
	$v_4 = \frac{6 + \sqrt{15}}{21},$	$v_5 = \frac{9 - 2\sqrt{15}}{21},$	$W_c = \frac{155 + \sqrt{15}}{1200}\Delta$				

The sampling point $\mathbf{r}' = \mathbf{r}_s$ of the global frame is obtained from (62) and (63).

TABLE II
EQUAL WEIGHT NINE POINT INTEGRATION FORMULA

Sample η_s, ξ_s	v_1, v_2	v_2, v_1	v_1, v_4	v_4, v_1	v_2, v_3	v_3, v_2	v_3, v_3	v_2, v_4	v_4, v_2
Weight	$W_s = \frac{\Delta}{9},$								
	$v_1 = \frac{1}{18},$	$v_2 = \frac{2}{9},$	$v_3 = \frac{7}{18},$	$v_4 = \frac{13}{18}$					

The sampling point $\mathbf{r} = \mathbf{r}_s$ of the global frame is given by (62) and (63).

ary ∂T . Obviously, when the source and the observation triangle are not the same, one may use numerical integration formulas of lower degree, thereby reducing the computational cost.

VI. CONCLUSION

Formulas for the static and dynamic potentials due to linearly varying source distributions defined on a planar triangle are presented. The singular kernels considered are the 3-D Green's function and its gradient, both for the static and the dynamic case. The expressions of the potential integrals involving the gradient of the 3-D Green's functions are valid also in the limit of observation point located in the source region; this permits one to simplify surface integral formulations, since use of these results implies not to perform the analytical extraction of the residue contributions in the development of the formulation.

In the static case the evaluation of potentials is performed analytically and compact form results are given in terms of readily identifiable geometrical quantities, which facilitates numerical implementation. In the dynamic case the results are given in a form suited for numerical integration.

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